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DAY 5 — DHT22 & LM35 Sensor Integration

Temperature Sensing with Arduino

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What is the DHT22?

Sensor Overview

The DHT22 is a digital temperature and humidity sensor that uses a single-wire protocol. It includes a capacitive humidity element and a thermistor for temperature — both readings packaged in one 4-pin module. No ADC required: the sensor outputs pre-converted digital values.

Temperature Range

−40°C to +80°C with ±0.5°C accuracy

Humidity Range

0–100% RH with ±2–5% accuracy

pinout.txt

```
DHT22 Quick Reference

Pin 1  → VCC (3.3–5.5V)
Pin 2  → DATA + 10kΩ pull-up
Pin 3  → NC (No Connect)
Pin 4  → GND

Output: Digital (pre-converted)
Library: Adafruit DHT sensor
```

💡 💡 DHT22 is digital — unlike LM35, it requires no ADC calculation. Library does all the math.

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Installing the DHT Library

Step 1: Library Manager

- In Arduino IDE, go to:
- Sketch → Include Library
- → Manage Libraries...
- Search: DHT sensor library

Step 2: Find & Install

- Find the library by Adafruit
- Click Install All when prompted
- This installs both:
- • DHT sensor library
- • Adafruit Unified Sensor

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Wiring DHT22 to Arduino

Connect the DHT22 using 4 pins. The DATA line MUST have a 10kΩ pull-up resistor to VCC — without it, the signal floats and reads fail randomly.

1

VCC

Connect Pin 1 to 5V rail on Arduino.

2

DATA + Pull-up

Connect Pin 2 to Arduino D2. Add 10kΩ between DATA and VCC.

3

GND

Connect Pin 4 to GND rail. Pin 3 (NC) is left unconnected.

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Reading Temperature & Humidity

Use the Adafruit DHT library. Create a DHT object with the pin and type, then call readTemperature() and readHumidity() each loop iteration.

temp.ino

```
#include <DHT.h>
#define DHTPIN 2  DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

void setup() {  dht.begin(); }

void loop() {

    float h = dht.readHumidity();
    float t = dht.readTemperature();
}
```

`#include <DHT.h>`

Include the Adafruit DHT library at top of sketch

`DHT dht(2, DHT22);`

Create sensor object: pin 2, type DHT22

`dht.readTemperature()`

Returns °C as float. Pass true for °F

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Handling Sensor Read Failures

What is NaN?

```
isnan(h || isnan(t)
```

NaN (Not a Number)	When DHT22 communication fails, the library returns IEEE 754 NaN. isnan() checks for this floating-point sentinel value.
Why Reads Fail	• Polling faster than 0.5 Hz (2s) • Missing 10kΩ pull-up • Power instability
Safe Pattern	Check isnan() before using values. Return early to skip the bad loop iteration.

isnan_check.ino

```
if (isnan(h) || isnan(t)) {
  Serial.println("Read failed!");
  return;  // skip loop
}
```

isnan() Notes

- Works on any float value
- NaN + any number = NaN (propagates)
- Always check before Serial.print()

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millis() vs delay() — Timing

Why Avoid delay()?

delay() freezes the entire CPU — no buttons, no sensors, no logic runs during the wait. millis() returns elapsed ms since boot, letting you check if enough time has passed without blocking.

`delay(ms)`

Blocks CPU for exact ms. Simple but freezes everything.

`millis()`

Returns unsigned long ms since boot. Non-blocking — CPU keeps running.

`unsigned long`

Variable type needed for millis(). Must be unsigned long to handle overflow.

timing.ino

```
unsigned long prev = 0;
const long interval = 2000;

void loop() {
  unsigned long now = millis();
  if (now - prev >= interval) {
    prev = now;
    // read sensor here
  }
}
```

💡 ⚠️ millis() overflows after ~49 days but subtraction still works correctly with unsigned long.

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Complete DHT22 Sketch

Full Non-Blocking Sketch

Combines all concepts: DHT library, sensor read, isnan() validation, millis() timing. This sketch reads temperature and humidity every 2 seconds without freezing the CPU.

dht22_complete.ino

```
#include <DHT.h>
#define DHTPIN 2  DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);
unsigned long prev = 0;

void setup() { Serial.begin(9600);
dht.begin(); }

void loop() {
  unsigned long now = millis();
  if (now - prev >= 2000) {
    prev = now;
    float h = dht.readHumidity();
    float t = dht.readTemperature();
    if (isnan(h)||isnan(t)) { return; }
    Serial.print("T:"); Serial.print(t,1);
Serial.println(" C");
  } }
```

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What is the LM35?

Analog Temperature Sensor

The LM35 is a precision analog temperature sensor with a linear output of 10mV per °C. Unlike DHT22, it uses the Arduino's built-in ADC — no library needed. Simple, reliable, and highly accurate.

Output Voltage

10mV per °C (e.g., 25°C → 250mV output)

Temp Range

−55°C to +150°C, ±0.5°C accuracy

Power & Interface

4–30V supply, analog output — connect to A0

lm35_pinout.txt

LM35 Quick Reference

```
Pin 1  → +VS (4–30V supply)
Pin 2  → VOUT → Arduino A0
Pin 3  → GND

Formula:
Vout = Temp_C * 0.01V
No library required!
```

💡 💡 LM35 vs DHT22: LM35 = temperature only (analog). DHT22 = temp + humidity (digital).

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Wiring LM35 to Arduino

3-Pin Analog Connection

The LM35 has just 3 pins: power, output, and ground. No pull-up resistor needed. Connect the output directly to any analog input pin (A0–A5).

Pin 1 → 5V

Power supply — connect to Arduino 5V rail

Pin 2 → A0

Analog output — connect to Arduino analog pin A0

Pin 3 → GND

Ground — connect to Arduino GND rail

lm35_wiring.txt

```
LM35 Wiring Summary

LM35 Pin 1 (+VS)
  → Arduino 5V
LM35 Pin 2 (VOUT)
  → Arduino A0 (analog)
LM35 Pin 3 (GND)
  → Arduino GND
```

💡 💡 LM35 wiring is simpler than DHT22 — no pull-up resistor, no library needed.

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Reading LM35 Temperature

ADC to Celsius Conversion

LM35 outputs 10mV/°C. Arduino's 10-bit ADC maps 0–5V to 0–1023. To get Celsius: multiply ADC value by (5.0/1024.0) to get voltage, then multiply by 100 to get °C.

`analogRead(A0)`

Returns 0–1023 (10-bit ADC value from LM35 output)

`* (5.0 / 1024.0)`

Converts ADC count to voltage (0–5V range)

`* 100.0`

Converts voltage to °C (10mV per degree = 100 × V)

lm35_read.ino

```
void setup() {
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(A0);
  float tempC = raw * (5.0/1024.0) *
100.0;
  Serial.print("Temp: ");
  Serial.println(tempC, 1);
  delay(1000);
}
```

💡 💡 Formula: tempC = analogRead(A0) * (5.0 / 1024.0) * 100.0

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Day 5 — Key Takeaways

What You Learned Today

Today covered two temperature sensors: the digital DHT22 (temp + humidity via library) and the analog LM35 (temp only via ADC formula). Both are connected to Arduino and output to Serial Monitor.

DHT22 Library

Install Adafruit DHT + Unified Sensor via Library Manager

isnan() + millis()

Always validate reads; use millis() for non-blocking timing

LM35 Formula

tempC = analogRead(A0) * (5.0 / 1024.0) * 100.0

summary.txt

```
DHT22 vs LM35 Summary

DHT22  Digital, 1-Wire
  Temp + Humidity
  Needs: Adafruit library

LM35   Analog, 3-Pin
  Temp only (no humidity)
  Needs: analogRead + math
```

💡 💡 DHT22 = digital, needs library. LM35 = analog, needs ADC math. Both output °C to Serial.

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DHT22 vs LM35 — Comparison

Choosing the Right Sensor

Both sensors measure temperature, but they differ in interface, accuracy, and capabilities. Use this comparison to decide which sensor fits your project requirements.

DHT22 Advantages
Measures humidity too; digital output needs no ADC math; more accurate at ±0.5°C

LM35 Advantages
Simpler wiring (no pull-up); no library needed; faster response; wider temp range

Use DHT22 when…
You need humidity data or want a library-based workflow

comparison.txt

Feature Comparison

Feature	DHT22	LM35
Output	Digital	Analog
Humidity	Yes	No
Library	Required	Not needed
Pull-up R	10k needed	No
Temp Range	-40~+80C	-55~+150C
Accuracy	±0.5C	±0.5C
Pins	4	3

💡 💡 Use LM35 when: temperature-only is enough. Use DHT22 when: you also need humidity.

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Thank You

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